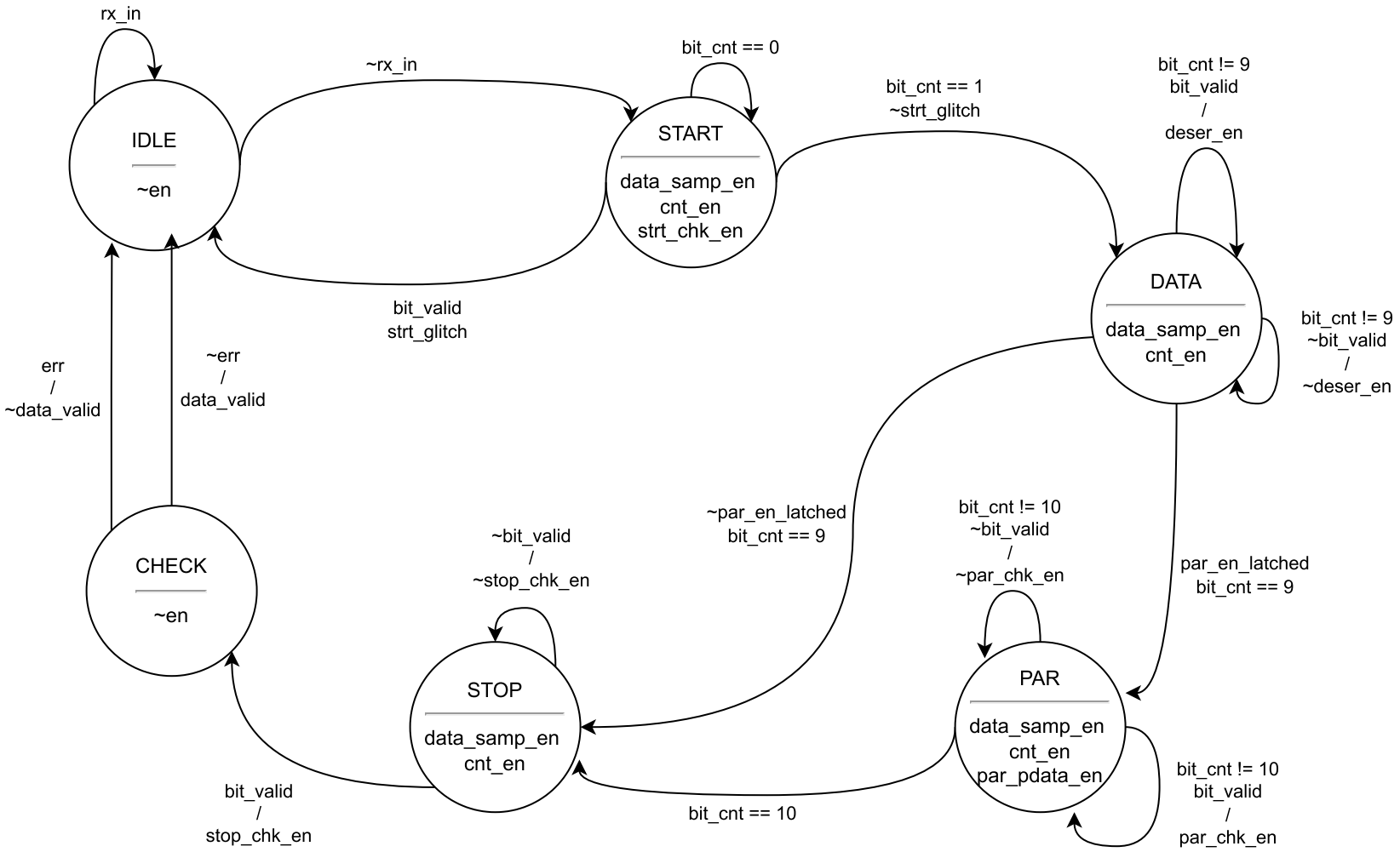
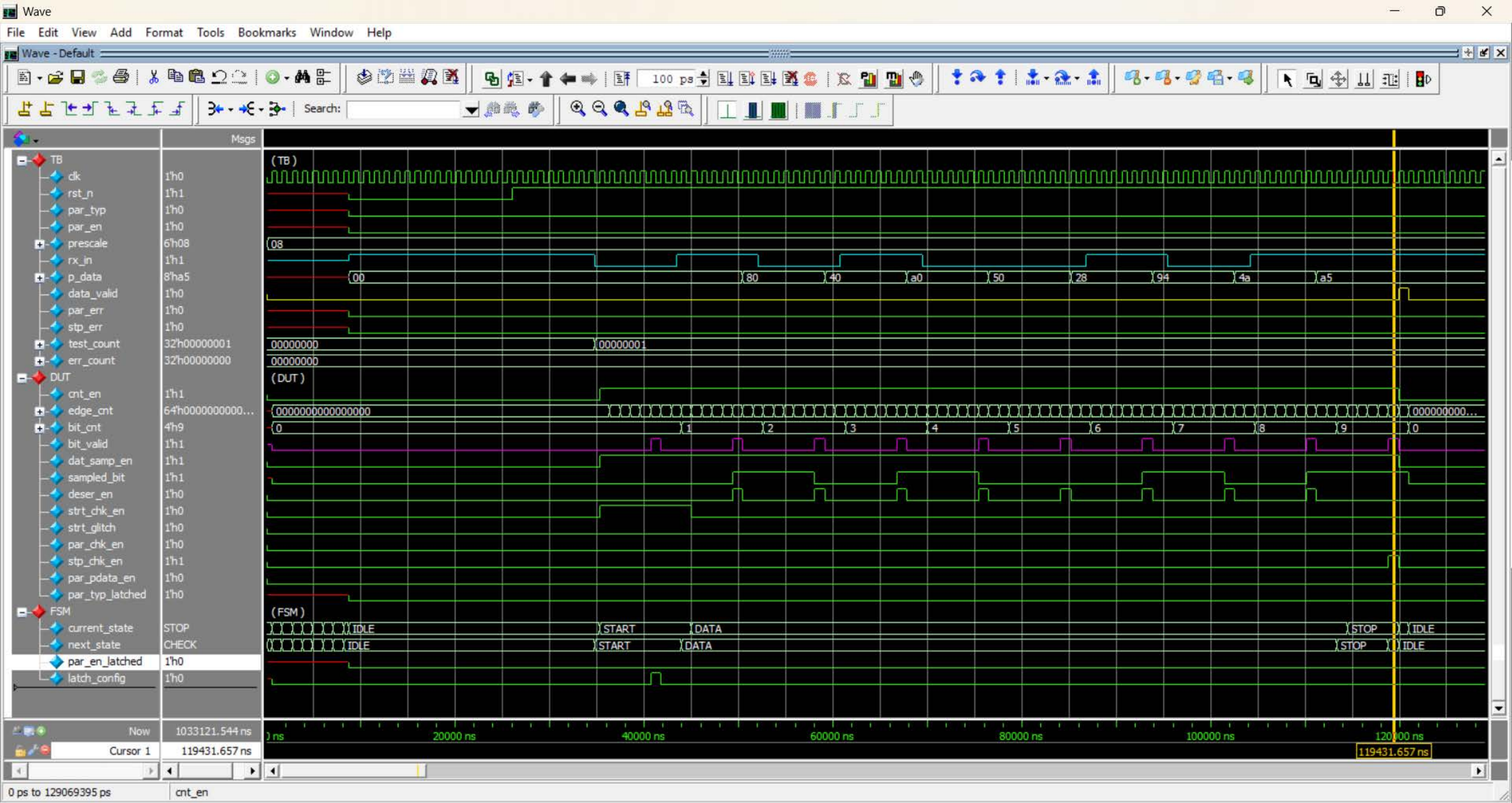
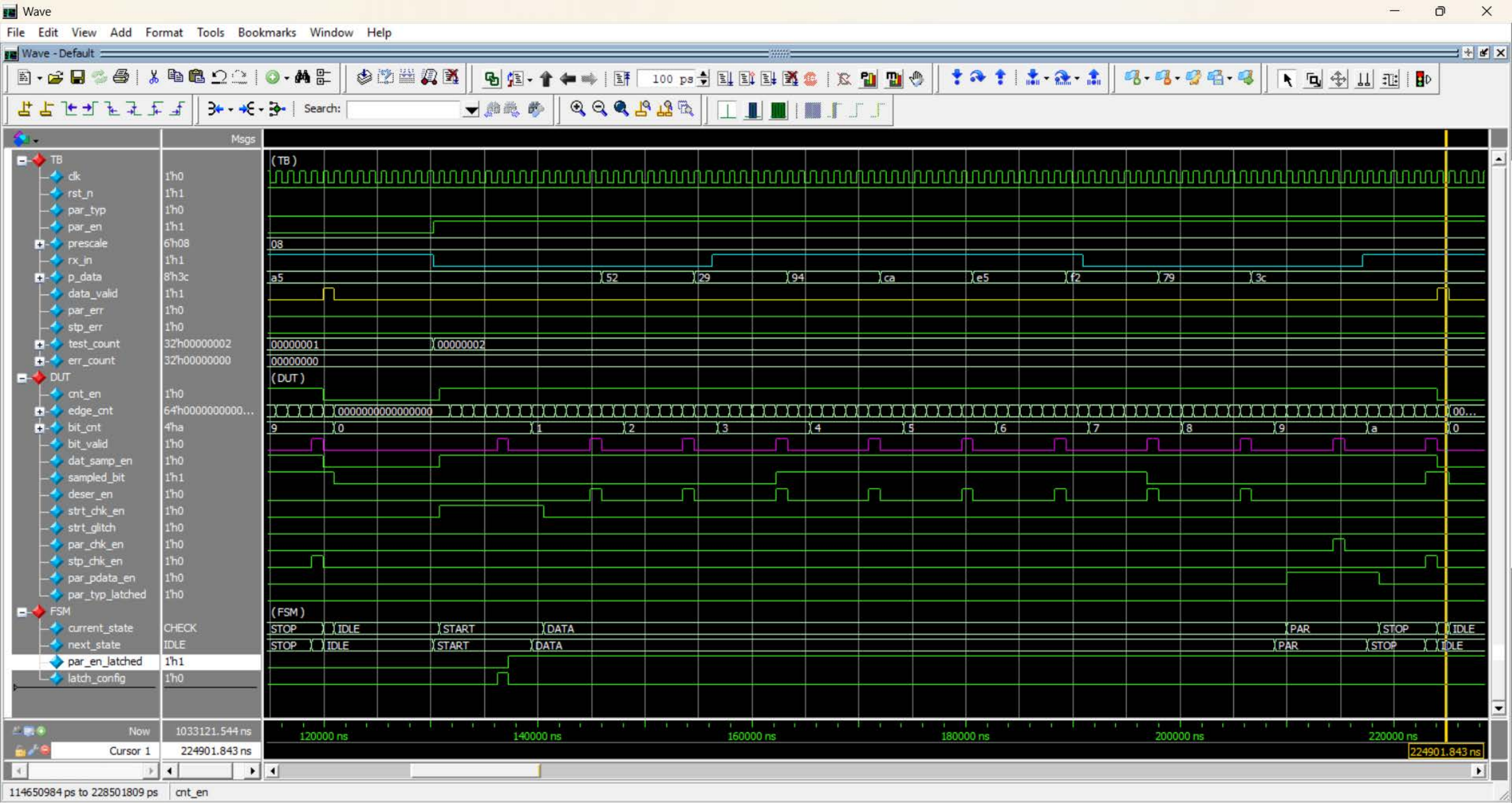
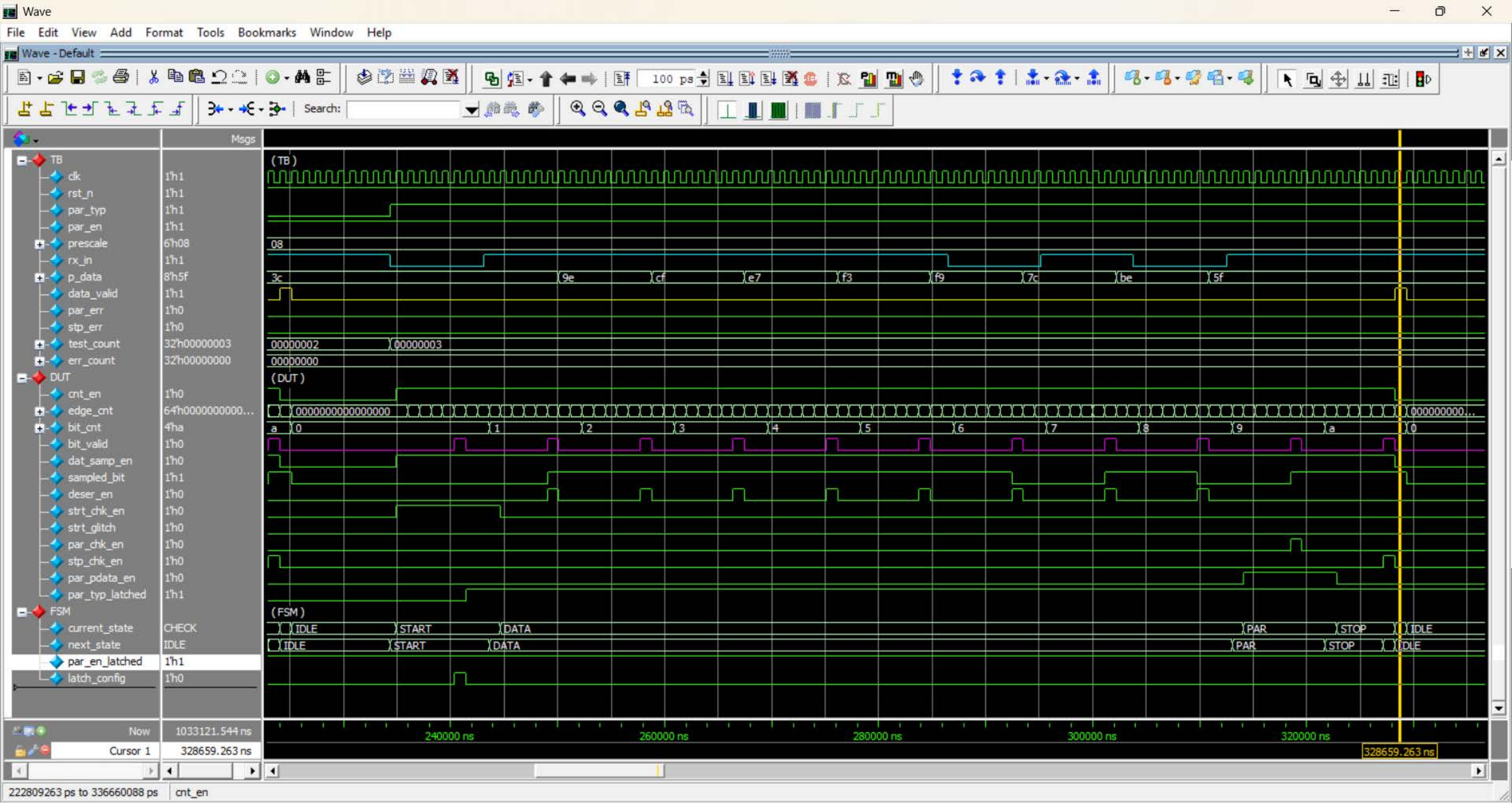
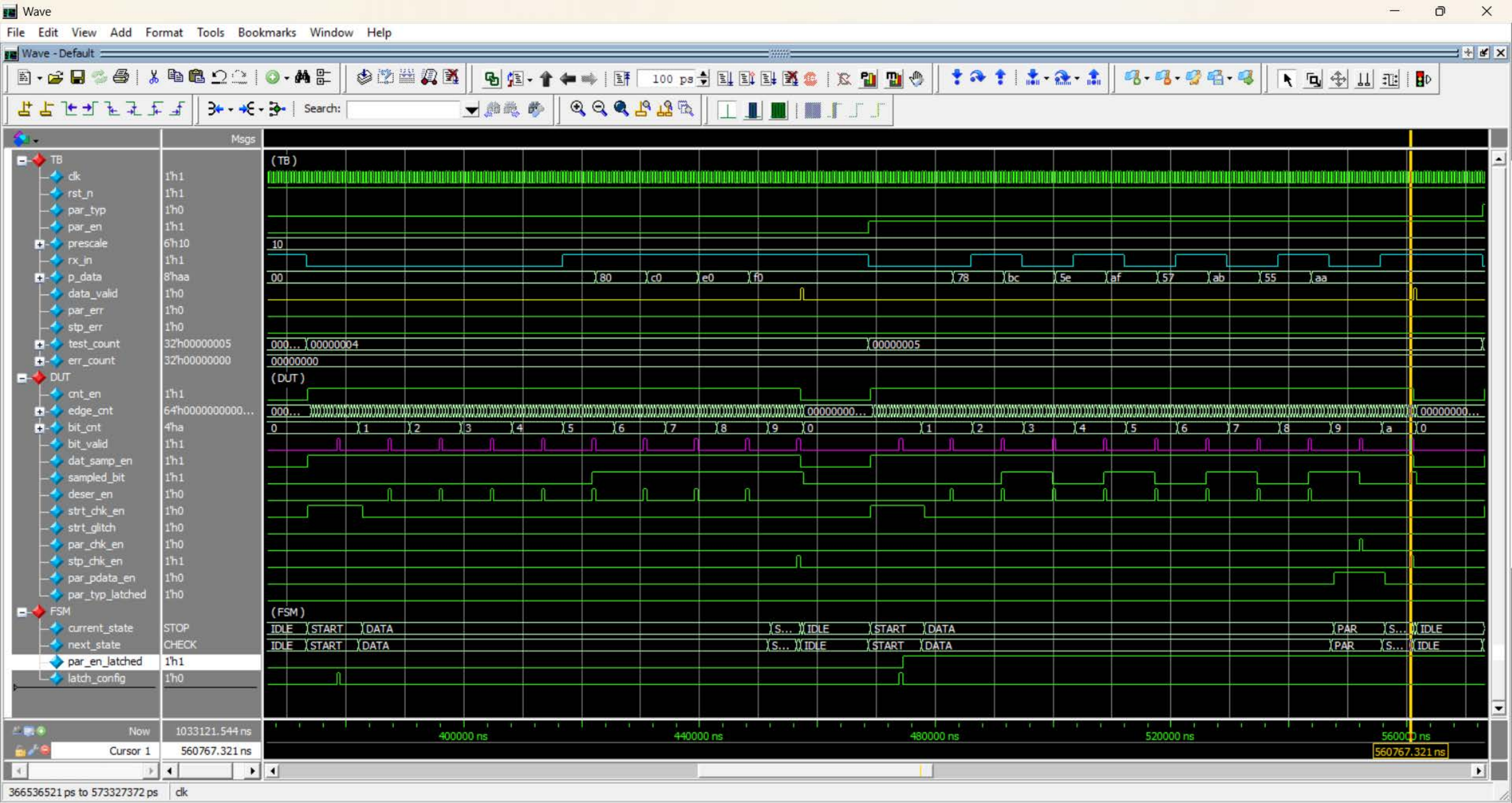
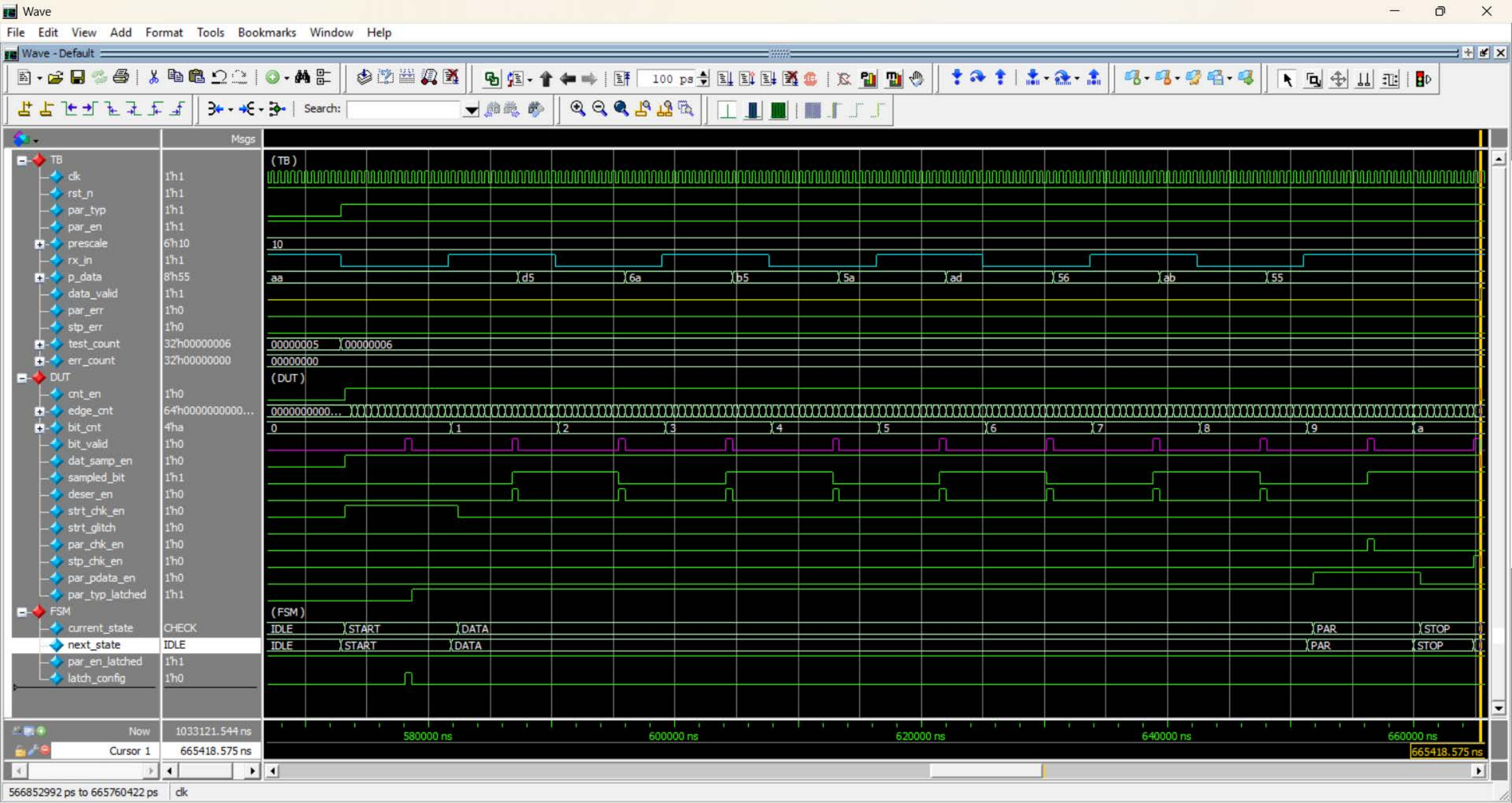

rx_in
IDLE
~en
~rx_in
START
data_samp_en
cnt_en
strt_chk_en
bit_cnt == 0
bit_cnt == 1
~strt_glitch
bit_valid
strt_glitch
DATA
data_samp_en
cnt_en
bit_cnt != 9
bit_valid
/
deser_en
bit_cnt != 9
~bit_valid
/
~deser_en
err
/
~data_valid
~err
/
data_valid
CHECK
~en
~par_en_latched
bit_cnt == 9
~bit_valid
/
~stop_chk_en
STOP
data_samp_en
cnt_en
bit_cnt != 10
~bit_valid
/
~par_chk_en
PAR
data_samp_en
cnt_en
par_pdata_en
par_en_latched
bit_cnt == 9
bit_cnt != 10
bit_valid
/
par_chk_en
bit_cnt == 10
bit_valid
/
stop_chk_en

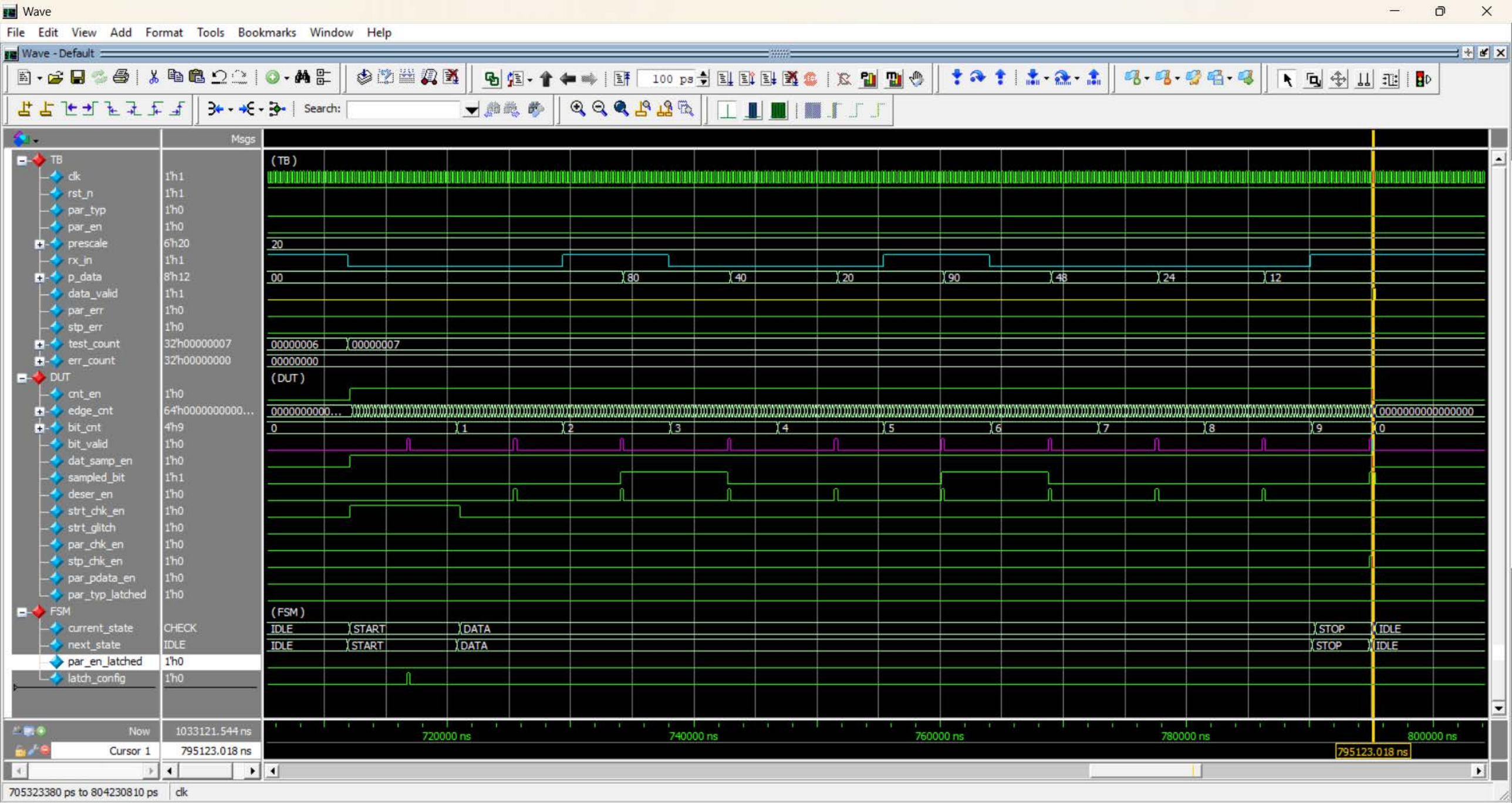

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h0
rst_n 1'h1
par_typ 1'h0
par_en 1'h0
prescale 6'h08
rx_in 1'h1
p_data 8'ha5
data_valid 1'h0
par_err 1'h0
stp_err 1'h0
test_count 32'h00000001
err_count 32'h00000000
DUT
cnt_en 1'h1
edge_cnt 64'h0000000000...
bit_cnt 4'h9
bit_valid 1'h1
dat_samp_en 1'h1
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h1
par_pdata_en 1'h0
par_typ_latched 1'h0
FSM
current_state STOP
next_state CHECK
par_en_latched 1'h0
latch_config 1'h0
(TB)
08
00 80 40 a0 50 28 94 4a a5
00000000 00000001
00000000
(DUT)
0000000000000000
0 1 2 3 4 5 6 7 8 9 0
(FSM)
IDLE START DATA STOP IDLE
IDLE START DATA STOP IDLE
Now 1033121.544 ns
Cursor 1 119431.657 ns
20000 ns 40000 ns 60000 ns 80000 ns 100000 ns 120000 ns
119431.657 ns
0 ps to 129069395 ps
cnt_en

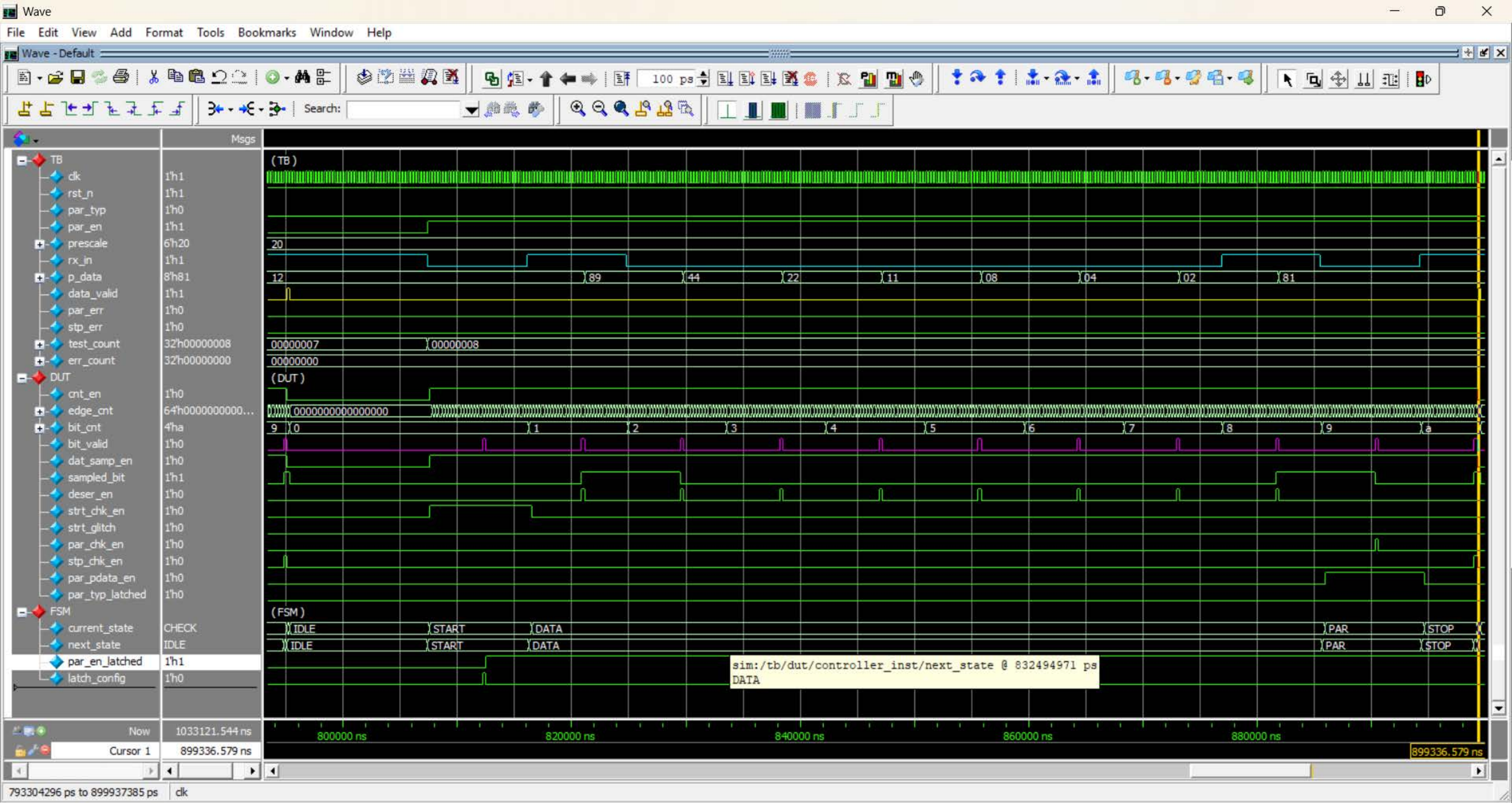

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h0
rst_n 1'h1
par_typ 1'h0
par_en 1'h1
prescale 6'h08
rx_in 1'h1
p_data 8'h3c
data_valid 1'h1
par_err 1'h0
stp_err 1'h0
test_count 32'h00000002
err_count 32'h00000000
DUT
cnt_en 1'h0
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h0
dat_samp_en 1'h0
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h0
par_pdata_en 1'h0
par_typ_latched 1'h0
FSM
current_state CHECK
next_state IDLE
par_en_latched 1'h1
latch_config 1'h0
(TB)
08
a5 52 29 94 ca e5 f2 79 3c
00000001 00000002
00000000
(DUT)
0000000000000000
9 0 1 2 3 4 5 6 7 8 9 a 0
(FSM)
STOP IDLE START DATA PAR STOP IDLE
STOP IDLE START DATA PAR STOP IDLE
Now 1033121.544 ns
Cursor 1 224901.843 ns
120000 ns 140000 ns 160000 ns 180000 ns 200000 ns 220000 ns
224901.843 ns
114650984 ps to 228501809 ps
cnt_en

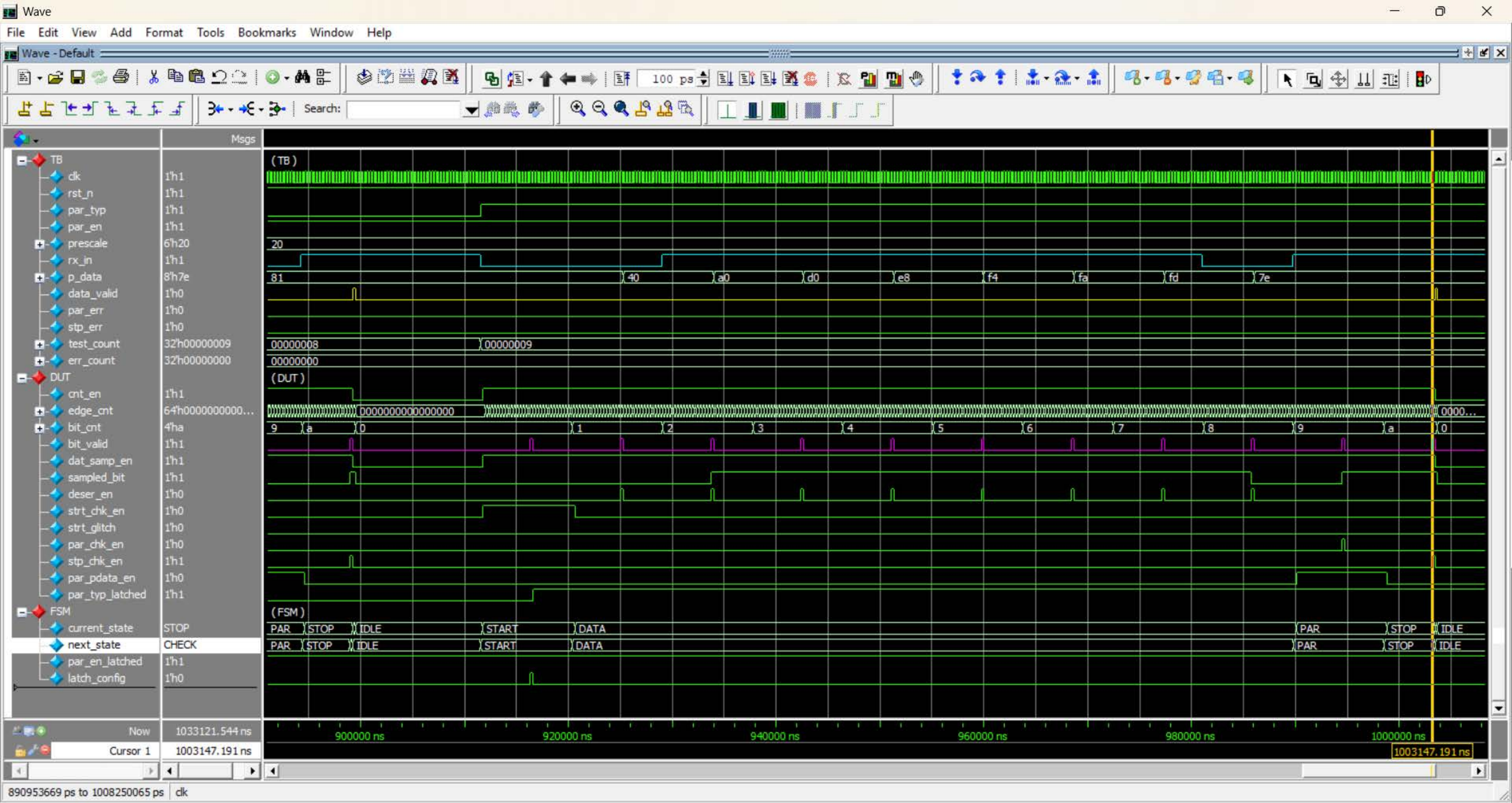

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h1
par_en 1'h1
prescale 6'h08
rx_in 1'h1
p_data 8'h5f
data_valid 1'h1
par_err 1'h0
stp_err 1'h0
test_count 32'h00000003
err_count 32'h00000000
DUT
cnt_en 1'h0
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h0
dat_samp_en 1'h0
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h0
par_pdata_en 1'h0
par_typ_latched 1'h1
FSM
current_state CHECK
next_state IDLE
par_en_latched 1'h1
latch_config 1'h0
(TB)
08
3c 9e cf e7 f3 f9 7c be 5f
00000002 00000003
00000000
(DUT)
0000000000000000
a 0 1 2 3 4 5 6 7 8 9 a 0
(FSM)
IDLE START DATA PAR STOP IDLE
IDLE START DATA PAR STOP IDLE
Now 1033121.544 ns
Cursor 1 328659.263 ns
240000 ns 260000 ns 280000 ns 300000 ns 320000 ns
328659.263 ns
222809263 ps to 336660088 ps
cnt_en

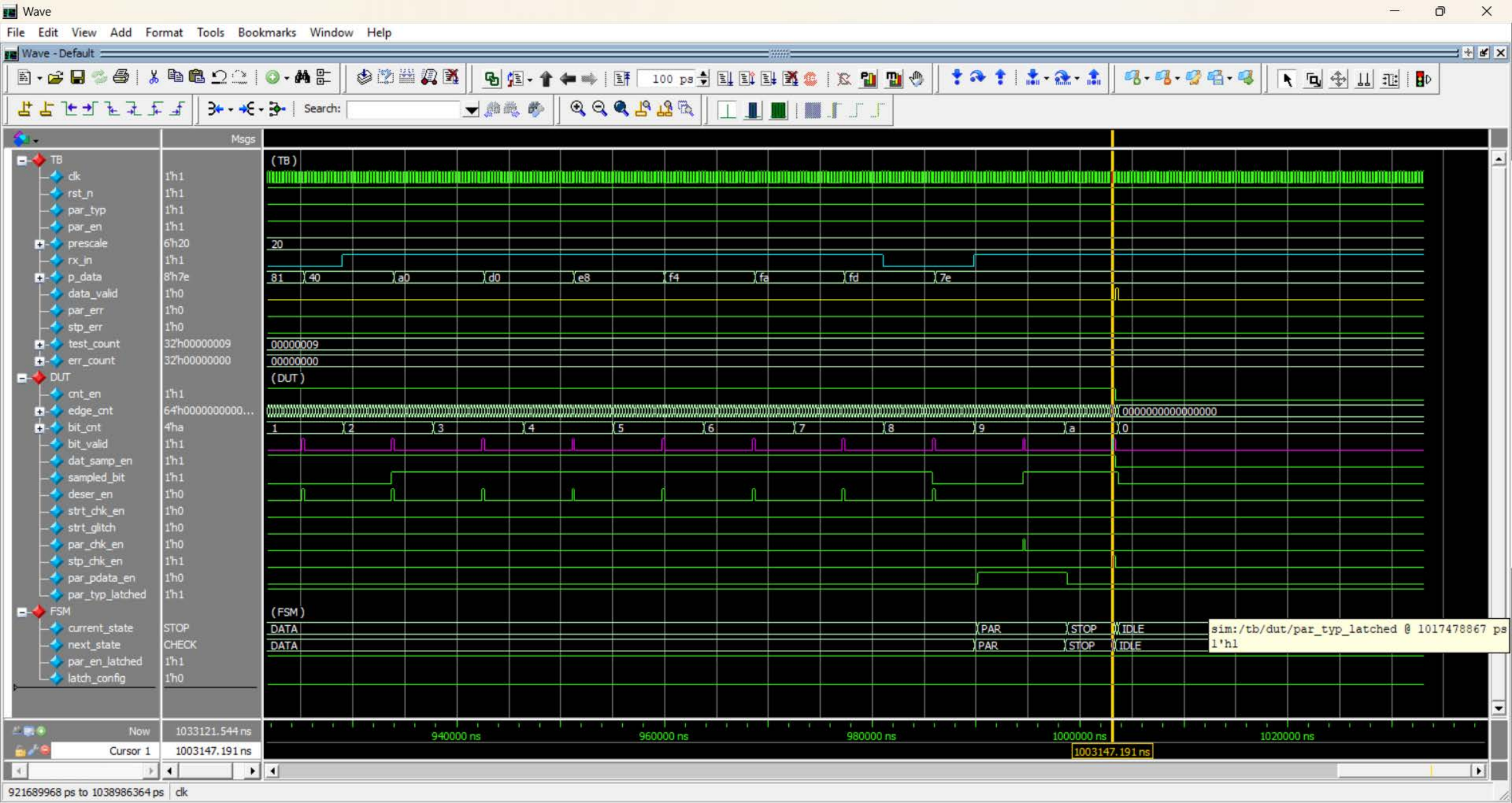

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h0
par_en 1'h1
prescale 6'h10
rx_in 1'h1
p_data 8'haa
data_valid 1'h0
par_err 1'h0
stp_err 1'h0
test_count 32'h00000005
err_count 32'h00000000
DUT
cnt_en 1'h1
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h1
dat_samp_en 1'h1
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h1
par_pdata_en 1'h0
par_typ_latched 1'h0
FSM
current_state STOP
next_state CHECK
par_en_latched 1'h1
latch_config 1'h0
(TB)
(DUT)
(FSM)
00 80 c0 e0 f0 78 bc 5e af 57 ab 55 aa
IDLE START DATA S... IDLE START DATA PAR S... IDLE
Now 1033121.544 ns
Cursor 1 560767.321 ns
400000 ns 440000 ns 480000 ns 520000 ns 560000 ns
560767.321 ns
366536521 ps to 573327372 ps clk

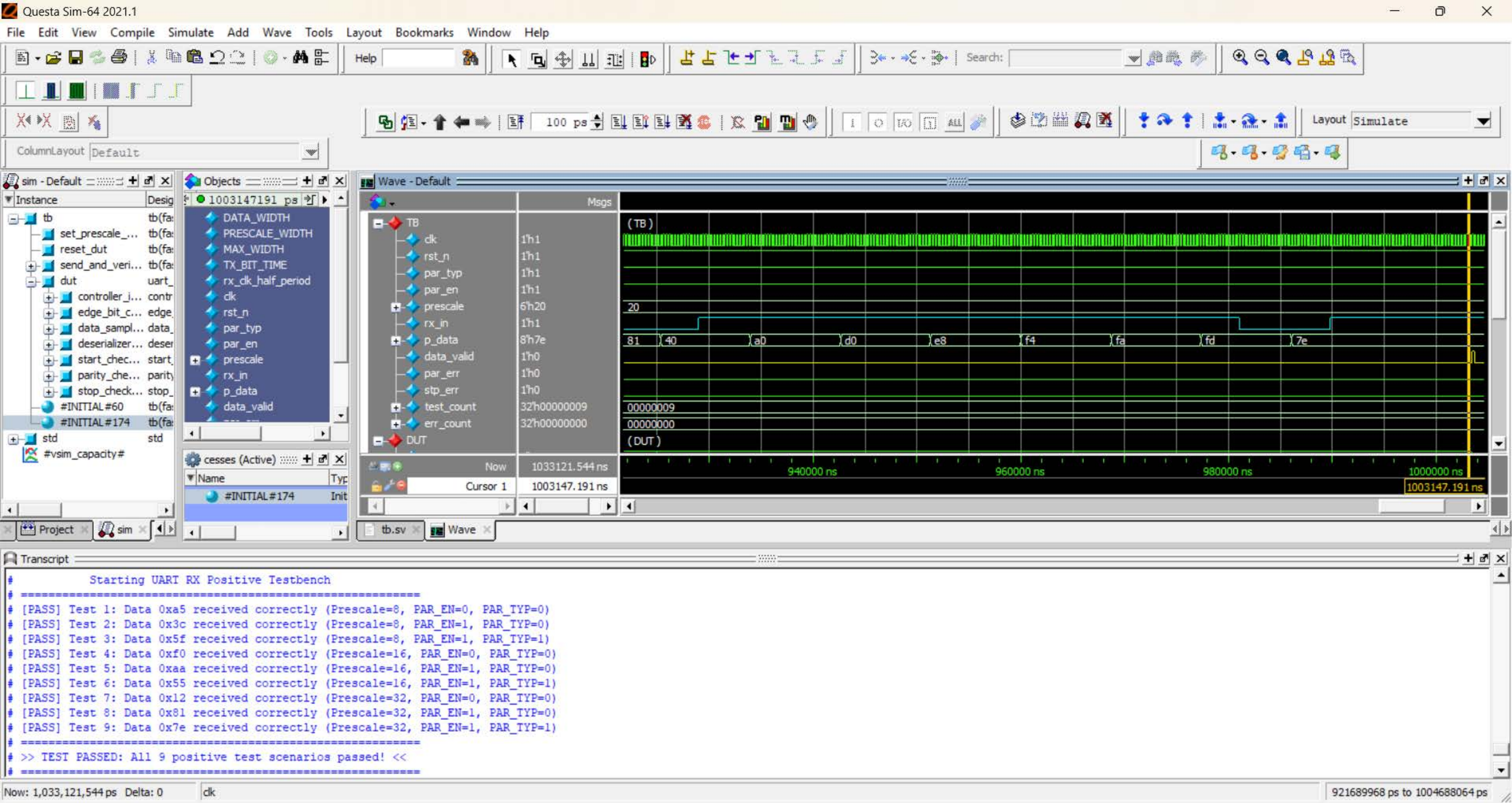

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h1
par_en 1'h1
prescale 6'h10
rx_in 1'h1
p_data 8'h55
data_valid 1'h1
par_err 1'h0
stp_err 1'h0
test_count 32'h00000006
err_count 32'h00000000
DUT
cnt_en 1'h0
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h0
dat_samp_en 1'h0
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h0
par_pdata_en 1'h0
par_typ_latched 1'h1
FSM
current_state CHECK
next_state IDLE
par_en_latched 1'h1
latch_config 1'h0
(TB)
10
aa d5 6a b5 5a ad 56 ab 55
00000005 00000006
00000000
(DUT)
0000000000...
0 1 2 3 4 5 6 7 8 9 a
(FSM)
IDLE START DATA PAR STOP
IDLE START DATA PAR STOP
Now 1033121.544 ns
Cursor 1 665418.575 ns
580000 ns 600000 ns 620000 ns 640000 ns 660000 ns
665418.575 ns
566852992 ps to 665760422 ps clk


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h0
par_en 1'h0
prescale 6'h20
rx_in 1'h1
p_data 8'h12
data_valid 1'h1
par_err 1'h0
stp_err 1'h0
test_count 32'h00000007
err_count 32'h00000000
DUT
cnt_en 1'h0
edge_cnt 64'h0000000000...
bit_cnt 4'h9
bit_valid 1'h0
dat_samp_en 1'h0
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h0
par_pdata_en 1'h0
par_typ_latched 1'h0
FSM
current_state CHECK
next_state IDLE
par_en_latched 1'h0
latch_config 1'h0
(TB)
20
00 80 40 20 90 48 24 12
00000006 00000007
00000000
(DUT)
0000000000...
0 1 2 3 4 5 6 7 8 9 0
(FSM)
IDLE START DATA STOP IDLE
IDLE START DATA STOP IDLE
Now 1033121.544 ns
Cursor 1 795123.018 ns
720000 ns 740000 ns 760000 ns 780000 ns 800000 ns
795123.018 ns
705323380 ps to 804230810 ps clk


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h0
par_en 1'h1
prescale 6'h20
rx_in 1'h1
p_data 8'h81
data_valid 1'h1
par_err 1'h0
stp_err 1'h0
test_count 32'h00000008
err_count 32'h00000000
DUT
cnt_en 1'h0
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h0
dat_samp_en 1'h0
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h0
par_pdata_en 1'h0
par_typ_latched 1'h0
FSM
current_state CHECK
next_state IDLE
par_en_latched 1'h1
latch_config 1'h0
(TB)
20
12 89 44 22 11 08 04 02 81
00000007 00000008
00000000
(DUT)
0000000000000000
9 0 1 2 3 4 5 6 7 8 9 a
(FSM)
IDLE START DATA PAR STOP
IDLE START DATA PAR STOP
sim:/tb/dut/controller_inst/next_state @ 832494971 ps
DATA
Now 1033121.544 ns
Cursor 1 899336.579 ns
800000 ns 820000 ns 840000 ns 860000 ns 880000 ns
899336.579 ns
793304296 ps to 899937385 ps clk


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h1
par_en 1'h1
prescale 6'h20
rx_in 1'h1
p_data 8'h7e
data_valid 1'h0
par_err 1'h0
stp_err 1'h0
test_count 32'h00000009
err_count 32'h00000000
DUT
cnt_en 1'h1
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h1
dat_samp_en 1'h1
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h1
par_pdata_en 1'h0
par_typ_latched 1'h1
FSM
current_state STOP
next_state CHECK
par_en_latched 1'h1
latch_config 1'h0
(TB)
20
81
40
a0
d0
e8
f4
fa
fd
7e
00000008
00000009
00000000
(DUT)
0000000000000000
0000...
9
a
0
1
2
3
4
5
6
7
8
9
a
0
(FSM)
PAR STOP IDLE START DATA
PAR STOP IDLE START DATA
PAR STOP IDLE
PAR STOP IDLE
Now 1033121.544 ns
Cursor 1 1003147.191 ns
900000 ns
920000 ns
940000 ns
960000 ns
980000 ns
1000000 ns
1003147.191 ns
890953669 ps to 1008250065 ps
clk


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
TB
clk 1'h1
rst_n 1'h1
par_typ 1'h1
par_en 1'h1
prescale 6'h20
rx_in 1'h1
p_data 8'h7e
data_valid 1'h0
par_err 1'h0
stp_err 1'h0
test_count 32'h00000009
err_count 32'h00000000
DUT
cnt_en 1'h1
edge_cnt 64'h0000000000...
bit_cnt 4'ha
bit_valid 1'h1
dat_samp_en 1'h1
sampled_bit 1'h1
deser_en 1'h0
strt_chk_en 1'h0
strt_glitch 1'h0
par_chk_en 1'h0
stp_chk_en 1'h1
par_pdata_en 1'h0
par_typ_latched 1'h1
FSM
current_state STOP
next_state CHECK
par_en_latched 1'h1
latch_config 1'h0
(TB)
20
81 40 a0 d0 e8 f4 fa fd 7e
00000009
00000000
(DUT)
0000000000000000
1 2 3 4 5 6 7 8 9 a 0
(FSM)
DATA PAR STOP IDLE
DATA PAR STOP IDLE
sim:/tb/dut/par_typ_latched @ 1017478867 ps
1'h1
Now 1033121.544 ns
Cursor 1 1003147.191 ns
940000 ns 960000 ns 980000 ns 1000000 ns 1020000 ns
1003147.191 ns
921689968 ps to 1038986364 ps clk


Questa Sim-64 2021.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance
tb
set_prescale_...
reset_dut
send_and_veri...
dut
controller_i...
edge_bit_c...
data_sampl...
deserializer...
start_chec...
parity_che...
stop_check...
#INITIAL#60
#INITIAL#174
std
#vsim_capacity#
Objects
1003147191 ps
DATA_WIDTH
PRESCALE_WIDTH
MAX_WIDTH
TX_BIT_TIME
rx_clk_half_period
clk
rst_n
par_typ
par_en
prescale
rx_in
p_data
data_valid
cesses (Active)
Name
#INITIAL#174
Wave - Default
Msgs
TB
clk
rst_n
par_typ
par_en
prescale
rx_in
p_data
data_valid
par_err
stp_err
test_count
err_count
DUT
Now
1033121.544 ns
Cursor 1
1003147.191 ns
(TB)
20
81
40
a0
d0
e8
f4
fa
fd
7e
00000009
00000000
(DUT)
940000 ns
960000 ns
980000 ns
1000000 ns
1003147.191 ns
Project
sim
tb.sv
Wave
Transcript
Starting UART RX Positive Testbench
[PASS] Test 1: Data 0xa5 received correctly (Prescale=8, PAR_EN=0, PAR_TYP=0)
[PASS] Test 2: Data 0x3c received correctly (Prescale=8, PAR_EN=1, PAR_TYP=0)
[PASS] Test 3: Data 0x5f received correctly (Prescale=8, PAR_EN=1, PAR_TYP=1)
[PASS] Test 4: Data 0xf0 received correctly (Prescale=16, PAR_EN=0, PAR_TYP=0)
[PASS] Test 5: Data 0xaa received correctly (Prescale=16, PAR_EN=1, PAR_TYP=0)
[PASS] Test 6: Data 0x55 received correctly (Prescale=16, PAR_EN=1, PAR_TYP=1)
[PASS] Test 7: Data 0x12 received correctly (Prescale=32, PAR_EN=0, PAR_TYP=0)
[PASS] Test 8: Data 0x81 received correctly (Prescale=32, PAR_EN=1, PAR_TYP=0)
[PASS] Test 9: Data 0x7e received correctly (Prescale=32, PAR_EN=1, PAR_TYP=1)
>> TEST PASSED: All 9 positive test scenarios passed! <<
Now: 1,033,121,544 ps Delta: 0
clk
921689968 ps to 1004688064 ps